

Testing the Substitution Predictions of Demand Models: Evidence from the Blue Bell Recall

Wantian Huang*
University of Maryland

Job Market Paper

October 2026

Abstract

Demand models' substitution predictions inform merger review and the value of product variety. Using the 2015 withdrawal of the regional ice cream brand Blue Bell, I test how well models estimated on data through March 2015 predict the amount of demand retained by rivals and its allocation among them. Relative to markets Blue Bell never served, rivals recaptured about three quarters of its volume, and private label took half of that gain. Logit and random-coefficient models without nesting retain too little demand. A nested logit that pools private label with Blue Bell's mainstream tubs comes close both on how much demand stays with rivals and on which rivals receive it, and random-coefficient nested logit (RCNL), which allows consumer heterogeneity on the same nests, performs similarly. In this application, where private label sits in the nests matters more for these predictions than how much taste heterogeneity a model allows.

Paper available upon request: whuang11@umd.edu

*Department of Economics, University of Maryland, College Park, MD 20742; whuang11@umd.edu.